CPTS 475: Data Science
Project Report

Analysis of Frequent Locations and Movement Patterns

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Abstract: Geospatial analysis is the analysis of time-based data that corresponds to a location on Earth determined by its coordinates. This project aims to analyze an extensive dataset containing location records for 20 individuals over the course of multiple years to recognize patterns and identify frequently visited locations using clustering algorithms. A probabilistic model is trained to estimate a person's schedule, which can be used to predict a person's location based on the time and day of the week. By mapping out movement patterns we are able to find out where the individuals live as well as where their work locations are. Other frequently visited locations can also be observed, as well as trips, highlighting travel habits.

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I. Introduction

Geospatial data is time-based data that corresponds to a location on Earth determined by its coordinates [1]. Analysis of such data allows us to model the real world to be able to recognize behavioral trends and identify factors that affect movement of living entities as well as non-biological systems such as environmental, economic, and physical processes. Geospatial analysis is not a novel concept and has many applications in life sciences (such as geology and ecology), natural resource management, and national intelligence. It can also be used by companies to make business decisions [1]. Human movement patterns in particular are a major contributor to development and planning of urban areas, location of essential facilities and demand for transportation services [2]. Additionally, analysis of mobility in humans gives insights into the motivations for why humans tend to migrate, on a daily basis as well as throughout a person's lifetime. Migration is the foundation of human lifestyle, which led to the economic growth and cultural diversity that we observe today.

This project aims to analyze a dataset containing location ping records for 20 individuals over the course of multiple years to recognize patterns and identify frequently visited locations using DBSCAN clustering. The clusters are visualized in an interactive heatmap. With the frequent locations found we can estimate a person's home and work locations. From this, we can estimate a person's daily schedule, which can then be used to train a Naïve Bayes Classification model that can estimate a person's location based on the time and day of the week. With the current scope of this project, we are able to come to conclusions about how often people tend to travel over the course of many years, and how far they tend to venture from their home location. At a larger scale, the results of this project could be useful to utility companies to estimate energy usage in each neighborhood throughout the day, and also to urban planners and State Departments of Transportation (DOTs) to estimate traffic levels.

II. Problem Definition

The Locations dataset contains time stamped coordinate readings, which keeps track of a person's location on a minute basis over the period of multiple years. The extent of this dataset helps us recognize patterns and estimate the person's daily schedule and movement patterns. By using density-based clustering to group close by coordinates, we can distinguish a person's frequently visited locations. With this analysis, we find out the number of distinct locations a person visits in a month and distinguish their top five locations on a monthly basis. We can also estimate their home and work locations. By mapping this out, we can answer questions such as: Where are the individuals captured in this dataset located? Where do they work? Where are their frequently visited locations? How far from home do they tend to travel on any given day?

We can also use this analysis to create a model that predicts the category of the cluster a person is likely to be at a certain time of day. Two models are trained using a snippet of the Locations dataset, one with clients who reside in the Portland/Vancouver area and the other with clients who reside in Spokane. The predictive models can provide estimates of the location of a person who fits that profile. Training a model on the clustering results allows for the analysis of this project to be applied in a broader context, allowing for universal human movement analysis which can be extrapolated to the entire population of the area, instead of specific analysis of only the individuals captured in this dataset. Although the applications of this analysis are outside the scope of this project, understanding human mobility in this way and the conclusions made by this analysis are crucial to urban planning, traffic forecasting as well as observing the spread of viruses [3].

III. Algorithms and Models

The goal of this project is to implement clustering to identify frequent locations and train a classifier model to predict a person's location. Here are the algorithms best suited to achieve this task:

III.I. Frequent Location Clustering

The dataset is noisy and extensive, with location pings often being in the same coordinates or extremely close to one another, meaning the person is at the same geographical location. To handle this, the best course of action is to use a clustering algorithm. Clustering allows us to group location pings that are close together into clusters and identify high density areas where a lot of readings occur. For the purpose of this project, Density-Based Spatial Clustering of Applications (DBSCAN) clustering was chosen. This algorithm is appropriate for this project since it an unsupervised learning algorithm, which means it can group data points without prior specifications. This allows the algorithm to learn from the data, instead of setting the number of clusters beforehand. DBSCAN is able to form complex cluster shapes which can deal with the irregularity and randomness of location data, adapting to the persons schedule. Additionally, it is able to identify clusters of varying densities, allowing us to come to conclusions about frequent locations based on the density of points on the heatmap [4].

DBSCAN clustering works by classifying points as Core points, Border points and Noise points [Figure 1]. Core points have at least a minimum number of other points (MinPts) within a specified distance (epsilon) [4]. In our implementation, the optimal value of MinPts was found to be 3 while epsilon was 30m, with Euclidean distance. With a lower value of MinPts the clusters contained a lot of noise (locations where the individual did not spend a lot of time or rarely visited) while a higher value made the clusters too sparse and failed to capture locations where the individual spent less than 2 hours, meaning the clusters were overfitting to represent home/work locations. Epsilon was set to 30m to make sure points in the same location (for example, within a building) would fall into one cluster, without being too generalized.

Figure 1. DBSCAN Points Classification [4]

III.II. Location Predictive Model

To predict the person's location based on time and day, Naïve Bayes Classifier was used. This model is appropriate for this project since it can estimate the cluster the individual is most likely to be at, based on the conditional probabilities learnt from the dataset. The Naïve Bayes Classifier uses Bayes Theorem to classify data based on the probabilities of different classes given the features of the data [5]. Naïve Bayes assumes that each feature is independent meaning it is able to handle time-based features that are weakly correlated. Additionally, our goal is to predict categorical data using probabilistic time-based location data, which is well handled by this model. Moreover, this model can be evaluated for accuracy using a confusion matrix which shows how the model predictions compare to the actual results [6, 7].

IV. Implementation and Analysis

This project was implemented in Python, using libraries such as Pandas for data manipulation, Scikit-Learn for machine learning models and Folium for interactive visualizations. Since the objective is to observe patterns in movement for each client, the models were implemented separately, following the same procedure each time. The steps taken are described below

IV.I. Dataset Preprocessing

The Locations dataset was messy and extensive, so it was important to preprocess the data before clustering. To begin with, the raw data was imported into a data frame for easy processing. The schema of the data frame is detailed in Table 1. The raw data was cleaned and tidied as per convention. Missing values were handled by dropping rows for simplicity. Additionally, location reading pings were not always exact. To prevent noisy data, low accuracy data was excluded. This was determined by the *accuracy* field. The more accurate a reading is, the lower the margin of error, and so records with *accuracy* higher than 1500 were excluded. Data type conversions were also handled in this step.

Table 1. Locations data frame structure

Column Name	**Data Type**	**Description**
client_id	int64	Unique client identifier
datetime	datetime64[ns]	Date and timestamp of location reading
latitude	float64	Latitude coordinate of location reading
longitude	float64	Longitude coordinate of location reading
accuracy	int64	Expected difference between dataset reading and location reading, used to determine accuracy
month_year	period[M]	Extracted month and year from datetime
int_latitude	int64	Converted latitude to integer
int_longitude	Int64	Converted longitude to integer

IV.II. Exploratory Data Analysis

To explore the nuances of the dataset and gain a better understanding of the clients behavior and patterns, the data frame was grouped by *month_year* and sorted. First, the count of locations visited each month was found and displayed in a table [Figure 2]. Next, the data frame was grouped by *month_year*, *int_latitude* and *int_longitude* and then sorted by *count* to find the top five locations for each month. The results were displayed as a table [Figure 3].

```
==Number of Locations visited in a Month==

   month_year  count
0     2019-01   1256
1     2019-02  17063
2     2019-03  26081
3     2019-04  14202
4     2019-05   8114
..        ...    ...
58    2023-11   7361
59    2023-12   7529
60    2024-01   6800
61    2024-02   4629
62    2024-03   3403
```

Figure 2. Number of locations visited in a month by Client 182

```
==Most Visited Locations Each Month==

    month_year  int_latitude  int_longitude  count
0      2019-01            44           -121    950
1      2019-01            45           -122    282
2      2019-01            45           -121     24
3      2019-02            45           -122  11058
4      2019-02            45           -111   2822
..         ...           ...            ...    ...
253    2024-03            39           -106   1162
254    2024-03            38           -119    699
255    2024-03            38           -122    442
256    2024-03            38           -109    254
257    2024-03            38           -120    225
```

Figure 3. Most visited locations each month for Client 182

As seen in the outputs, the results of this were difficult to read and also contained a lot of repeated records, since each set of coordinates was treated as unique, even though they were physically close together. This led to the conclusion that a clustering algorithm would be the best method for this dataset.

IV.III. DBSCAN Clustering

DBSCAN clustering was used to find the locations where each client spent the most time per month. Due to the size of this dataset, it was required to downsize the dataset, as a consequence of the computation power needed to process the full dataset. So, the data was sorted into 30min interval time buckets and only the first ping for each interval was considered. Next, time spent at each bucket location was calculated. Before clustering, coordinates were projected into meters to be able to use Euclidean distance since geographic coordinates do not map equally to distance. Without this step, Haversine distance would need to be used for the clustering, which is computationally heavy. Lastly, the location data was grouped by month and clustered by DBSCAN using Euclidean distance, with epsilon as 30 and MinPts set to 3. Each cluster was also given a unique ID for identification. The centroids of each cluster were also found. To find the top locations for the client, the clustered buckets were grouped by month and sorted by time spent. The top five locations were compiled into a table, which can be seen in Figure 4.

```
=== TOP 5 LOCATIONS PER MONTH ===

       month cluster_uid  time_spent
0    2019-01   2019-01_1    390834.0
1    2019-01   2019-01_0     88144.0
2    2019-01   2019-01_2     26614.0
3    2019-02   2019-02_0   1218616.0
4    2019-02   2019-02_4    453787.0
..       ...         ...         ...
302  2024-03   2024-03_7    477214.0
303  2024-03   2024-03_2    458144.0
304  2024-03   2024-03_9     64674.0
305  2024-03   2024-03_0     63238.0
306  2024-03   2024-03_6     60994.0
```

Figure 4. Top 5 location cluster per month for Client 182

IV.IV. Frequent Location Analysis

Now that the location dataset is clustered, we can now analyze the top locations to find plausible home and work locations. To this *datetime* is parsed into *hour*, which is the time of day and *weekday* which is the day of the week. Then certain times of the day are categorized as "night", "workday" or "other". 9PM to 6AM is classified as nighttime, while 9AM to 5PM on a weekday is considered a workday. Any time outside of those time slots are categorized as "other". Each cluster in then labelled appropriately. Once each cluster is categorized, the home and work locations can be estimated. The cluster where most time is spent at night is considered the home cluster, while the cluster where most time is spent during the workday is considered the work cluster. The centroids of the home and work clusters are displayed as the output [Figure 5].

```
HOME coords: [  45.54950559 -122.6260189 ]
WORK coords: [  45.54948449 -122.62600635]
```

Figure 5. Home/Work clusters for Client 182

IV.V. Heatmap Visualization

For a clear representation of the DBSCAN clustering, a heatmap was used to visualize the clusters on an interactive geographical map. The clusters are also weighted by time spent. The density of the points is represented by a gradient, with red and yellow representing clusters where most time is spent while green and purple represent sparse gradients where not a lot of time was spent by the client. For ease of identification, home and work markers were added to the map. Arrows showing movement path are also displayed. The heatmaps are saved as an HTML files for easy viewing.

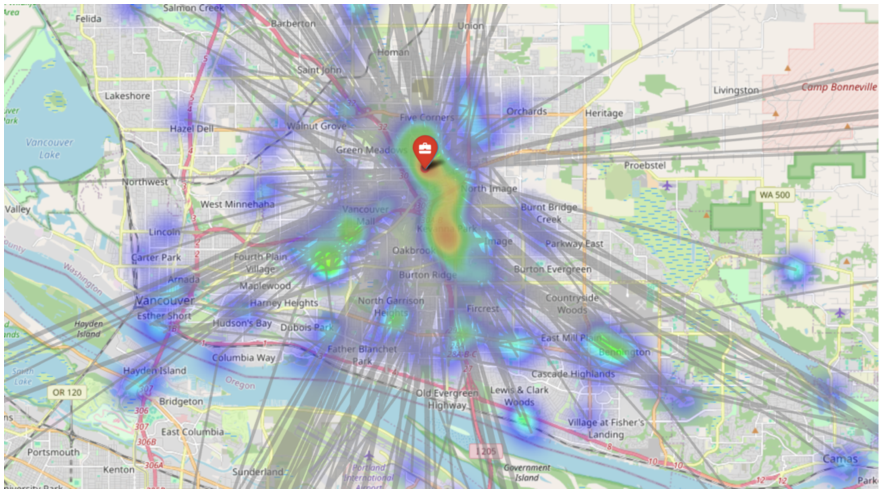

Figure 6. Showcase of Heatmap for Client 272

IV.VI. Predictive Model

With the generated heatmaps, it was observed that there were four individuals residing in the Portland/Vancouver area who work from home and five individuals living in the Spokane area. Two general predictive model were created, which predict the category of the cluster they are located at based on the time and day for each group of clients. Since this model aggregates multiple client datasets, it is implemented separately from the clustering done previously. The methodology for both models remain the same, only the selection of clients differs.

First, the data for each client is cleaned and filtered to remove missing values and drop inaccurate readings. To prevent noise, records outside the target city are excluded. For simplicity, KMeans clustering is used to identify common location clusters. Work and home clusters are identified as well. Each cluster is given a category label as “home”, “work” or “other” [Figure 7, Figure 8].

```
Model 1: Individuals living in Portland/Vancouver area
Client 64 Clustering:
Cluster labels:
Cluster 0: home, centroid = [  45.43644461 -122.71908843]
Cluster 1: other, centroid = [  47.73792617 -122.14440697]
Cluster 2: other, centroid = [  46.45098334 -122.91781453]
Cluster 3: work, centroid = [  45.52267577 -122.66873113]
Cluster 4: other, centroid = [  47.11076104 -122.64713628]
Cluster 5: other, centroid = [  47.44985084 -122.33449487]
Client 182 Clustering:
Cluster labels:
Cluster 0: other, centroid = [  45.54617209 -122.8073552 ]
Cluster 1: work, centroid = [  47.62005209 -122.05026724]
Cluster 2: home, centroid = [  45.54865288 -122.62546746]
Cluster 3: other, centroid = [  46.71721604 -122.86399766]
Cluster 4: other, centroid = [  45.40470565 -122.31557245]
Cluster 5: other, centroid = [  48.377279   -122.72426356]
Client 272 Clustering:
Cluster labels:
Cluster 0: home, centroid = [  45.66299727 -122.56810413]
Cluster 1: other, centroid = [  45.47485463 -122.67768628]
Cluster 2: other, centroid = [  47.61816314 -122.33881    ]
Cluster 3: other, centroid = [  45.58038923 -122.42594707]
Cluster 4: other, centroid = [  45.92233229 -122.41598419]
Cluster 5: work, centroid = [  45.69544846 -122.67296811]
Client 273 Clustering:
Cluster labels:
Cluster 0: work, centroid = [  45.56664688 -122.32897553]
Cluster 1: other, centroid = [  47.40624517 -122.50861287]
Cluster 2: home, centroid = [  45.63561288 -122.59528026]
Cluster 3: other, centroid = [  45.43298444 -122.80197821]
Cluster 4: other, centroid = [  46.68870341 -122.87812458]
Cluster 5: other, centroid = [  45.7708009  -122.54491108]
```

Figure 7. KMeans clustering for clients chosen for Model 1

```
Model 2: Individuals living in Spokane area
Client 67 Clustering:
Cluster labels:
Cluster 0: other, centroid = [  47.65228181 -117.40892563]
Cluster 1: work, centroid = [  47.59068189 -117.36310305]
Cluster 2: other, centroid = [  47.66746054 -117.34315915]
Cluster 3: other, centroid = [  47.66243222 -117.47799463]
Cluster 4: other, centroid = [  47.69050169 -117.41194994]
Cluster 5: home, centroid = [  47.6614691  -117.42792914]
Client 68 Clustering:
Cluster labels:
Cluster 0: home, centroid = [  47.67913438 -117.35284119]
Cluster 1: other, centroid = [  47.66407776 -117.42297214]
Cluster 2: other, centroid = [  47.43063316 -117.39040696]
Cluster 3: other, centroid = [  47.66255002 -117.3944565 ]
Cluster 4: other, centroid = [  47.64210187 -117.49123251]
Cluster 5: work, centroid = [  47.69476911 -117.37061213]
Client 70 Clustering:
Cluster labels:
Cluster 0: home, centroid = [  47.64293088 -117.32713956]
Cluster 1: work, centroid = [  47.66318098 -117.42838873]
Cluster 2: other, centroid = [  47.60968582 -117.36710067]
Cluster 3: other, centroid = [  47.6271968  -117.36933312]
Cluster 4: other, centroid = [  47.65244327 -117.40342224]
Cluster 5: other, centroid = [  47.65497737 -117.35887288]
Client 328 Clustering:
Cluster labels:
Cluster 0: home, centroid = [  47.43513309 -117.49686748]
Cluster 1: work, centroid = [  47.66378671 -117.42088226]
Cluster 2: other, centroid = [  47.60075496 -117.41326045]
Cluster 3: other, centroid = [  47.63810328 -117.35471102]
Cluster 4: other, centroid = [  47.44527217 -117.39346587]
Cluster 5: other, centroid = [  47.630427   -117.47230239]
Client 338 Clustering:
Cluster labels:
Cluster 0: home, centroid = [  47.64903356 -117.40950226]
Cluster 1: work, centroid = [  47.65378521 -117.31431757]
Cluster 2: other, centroid = [  47.66428957 -117.38017036]
Cluster 3: other, centroid = [  47.65931375 -117.3429923 ]
Cluster 4: other, centroid = [  47.65822414 -117.41756692]
Cluster 5: other, centroid = [  47.6873471 -117.4169939]
```

Figure 8. KMeans clustering for clients chosen for Model 2

Next, the labelled cluster points are concatenated to create a combined cluster cloud. Now for the predictive model, a Naïve Bayes Classifier is used, where P(state_label | hour, weekday). Which means that given a time and day, the model outputs the category of the cluster an individual who fits the profile of the clients considered in this model are most likely to be at. To this, a probability table is built with Laplace smoothing. The features of this model are *hour*, *weekday*, *is_weekend*, *is_workhour* and *is_sleephour* [Table 2]. The state_labels are *home*, *work*, and *other*. A probability table is then created which counts how many times each state occurs for each feature value. Laplace smoothing is used to make sure there is no zero probabilities. Then the conditional probabilities are calculated.

Table 2. Features used for Naïve Bayes classifier prediction

Feature	Definition
hour	The hour of the day, extracted from input
weekday	The day of the week, extracted from input
is_weekend	True if weekday is Saturday (5) or Sunday (6)
is_workhour	True if hour is between 9AM to 5PM
is_sleephour	True if hour is between 12AM to 5AM

Now the prediction function receives a datetime input and uses the probability table to predict the cluster category (state_label) based on the highest probability. To find the highest probability, Naïve Bayes multiplication is done, using the conditional probability for each feature, then the maximum value of that set determines the output of the prediction function. Lastly, the model accuracy is evaluated using a confusion matrix and other accuracy metrics such as precision and recall and client-wise accuracy.

V. Results and Discussion

The analysis provides us with two types of results: Heatmaps generated by DBSCAN clustering, and the model predictions with evaluation metrics.

V.I. DBSCAN Clustering Heatmaps

By clustering, we were able to make some general observations about the 20 individuals covered in the Locations dataset using the heatmap visualization . It was found that the most location points were in the person's home city, while the home location was the densest cluster. There were sparse clusters throughout the home city representing other locations the individual tends to frequent. With the movement arrows, we observe that movement begins and ends at the home location. Clustering corresponding to travel to other cities (for work or pleasure) can also be seen on the heatmap. With the tables found in the analysis report we can also view the top five frequent locations per month based on time spent in each cluster. Specific observations for each client can be found in Table 3.

Table 3. Indiviual analysis for each client

Client ID	City	Home Cluster Centroid	Work Cluster Centroid	WFH?	Additional Notes
64	Portland, OR	(45.44, -122.71)	(45.44, -122.71)	Yes	Tends to remain in PNW, with very few trips around the USA
67	Spokane, WA	(47.86, -117.46)	(47.86, -117.46)	Yes	Travels pretty often, mostly CA, CO, TX, DC
68	Spokane, WA	(47.68, -117.35)	(47.68, -117.35)	Yes	Tends to remain in WA, mainly travels between Spokane and Seattle
69	Atlanta, GA	(33.72, -84.41)	(33.72, -84.41)	Yes	Mostly remains in Atlanta, frequent trips to WA, DC
70	Spokane, WA	(47.64, -117.33)	(47.64, -117.33)	Yes	Tends to remain in WA, frequent trips to UT
175	Oakland, CA	(37.82, -122.27)	(37.82, -122.27)	Yes	Heavy traveler, frequent trips to LA and Seattle
177	Downers Grove, IL	(41.78, -88.043)	(41.78, -88.043)	Yes	Mainly remains in Chicago area, with frequent trips around the USA
179	Fort Worth, TX	(32.96, -97.09)	(32.96, -97.09)	Yes	Mostly remain in TX (Fort Worth/Houston), travels Phoenix and Seattle often
181	Mercer Island, WA	(47.54, -122.23)	(47.54, -122.23)	Yes	Mainly in Seattle area, with infrequent trips
182	Portland, OR	(45.55, -122.63)	(45.55, -122.63)	Yes	Often travels between Portland and Seattle, mainly travels on West USA
258	Bellefonte , PA	(40.92, -77.78)	(40.92, -77.78)	Yes	Mainly remains in PA, Travels frequently around the East coast

269	Cottage Lake, WA	(47.74, -122.11)	(47.74, -122.11)	Yes	Mainly remains in Seattle area, frequent travels to NY
272	Vancouver, WA	(45.67, -122.57)	(45.67, -122.57)	Yes	Remains in Vancouver/Portland area, rarely travels
273	Vancouver, WA	(45.65, -122.61)	(45.65, -122.61)	Yes	Mainly in Vancouver/Portland area, frequent trips to Seattle
276	Short Pump, VA	(37.64, -77.63)	(37.63, -77.64)	No	Commutes an office 5 minutes away from home, Frequent trips to Tallahassee as well as around USA
328	Spokane, WA	(47.48, -117.59)	(47.48, -117.59)	Yes	Tends to remain in WA, mainly travels between Spokane and Seattle
336	Redmond, WA	(47.65, -122.13)	(47.61, -122.20)	No	Commutes an office 11 minutes away from home, Mainly remains in Seattle area
338	Spokane, WA	(47.64, -117.23)	(47.65, -117.27)	No	Commutes an office 4 minutes away from home, Tends to remain in Spokane area, rare trips
343	Seattle, WA	(47.66, -122.33)	(47.66, -122.33)	Yes	Tends to remain in Seattle area, rare trips
344	Sunriver, OH	(43.88, -121.44)	(43.88, -121.44)	Yes	Mainly in PNW, but travels internationally often (Canada, Costa Rica, UK, Italy, Portugal)

We can now find answers to the exploratory questions defined before analysis.

Where are the individuals captured in this dataset located? The individuals live in various cities across the USA, although there are some who are located in the same city.
Where do they work? Most of the individuals work from home although there are a few who commute to an office location 5-10 minutes from their home (namely clients 276, 336, 338).

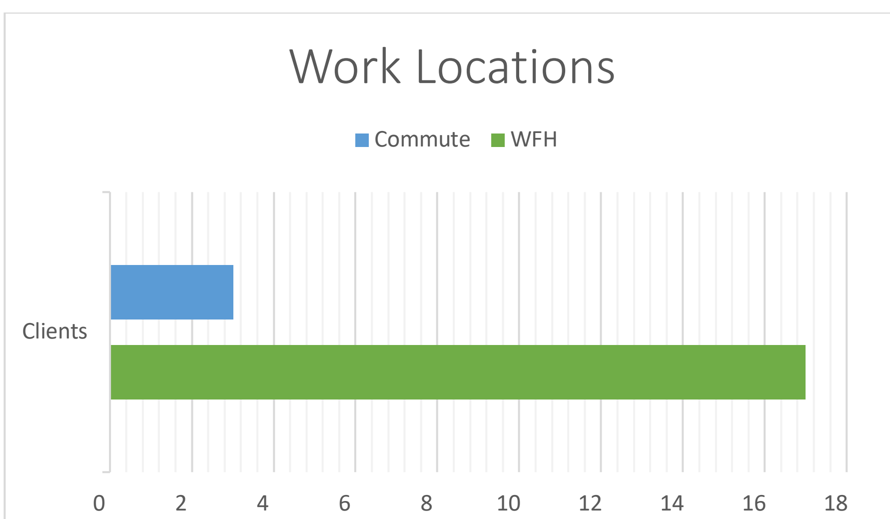

Figure 9. Bar chart showing work locations among the clients

Where are their frequently visited locations? These differ for each person but apart from their home and work, each person seems to have 3-5 places within their home city that they tend to frequent every month. This could be places like a mall, a hospital, a school, or restaurant.
How far from home do they tend to travel on any given day? On a daily basis, individuals either stay at home or travel within their home city. Movement patterns vary across a longer period of time. There are some who travel often to various places around the country or internationally, some who travel back and forth between nearby cities and others who are fairly sedentary and tend to remain in their home city with little to no travel.

The clustering approach used in this project seems to show accurate mobility behaviors of the clients, with clusters appearing on the map where the individual spent considerable time, along with the movement patterns showing travel habits. Although, based on the skewness of the data we can infer that the findings for work locations may not be accurate to real life. Since the home and work clusters are found by assuming that each person works in the standard 9-5 Monday-Friday schedule (or 1st shift) and finding the cluster where most time is spent during those hours of the day, this logic fails to consider individuals with an irregular work schedule or jobs during the 2nd or 3rd shift. The exact reasoning for this inaccuracy is hard to deduce due to the limitation of the dataset, which does not include personal information about the clients occupations.

V.II. Naïve Bayes Model Evaluation

The Naïve Bayes models receive an input of a date and time value and then use the probability table to predict the cluster label the individual is likely to be at [Figure 10, Figure 11].

```
Model Demonstration:
Datetime: 2024-11-18 04:30:00
Predicted category: home
Probabilities: {'home': np.float64(0.9992724217836755), 'work': np.float64(0.0005568728012538699), 'other': np.float64(0.0001707054150707287)}

Datetime: 2024-11-18 10:30:00
Predicted category: home
Probabilities: {'home': np.float64(0.999180805451242), 'work': np.float64(0.0006513456126153856), 'other': np.float64(0.0001678489361427559)}

Datetime: 2024-11-18 18:30:00
Predicted category: home
Probabilities: {'home': np.float64(0.9988869414329394), 'work': np.float64(0.0008726282486203373), 'other': np.float64(0.00024043031844010908)}
```

Figure 10. Model #1 Demonstration – Portland/Vancouver area

```
Model Demonstration:
Datetime: 2024-11-18 04:30:00
Predicted category: home
Probabilities: {'home': np.float64(0.9995598112436817), 'work': np.float64(0.0004042483980282457), 'other': np.float64(3.594035829009227e-05)}

Datetime: 2024-11-18 10:30:00
Predicted category: home
Probabilities: {'home': np.float64(0.9997231337230371), 'work': np.float64(0.00027566927825470956), 'other': np.float64(1.1969987081303657e-06)}

Datetime: 2024-11-18 18:30:00
Predicted category: home
Probabilities: {'home': np.float64(0.9984656977201937), 'work': np.float64(0.0013721400726110077), 'other': np.float64(0.00016216220719541814)}
```

Figure 11. Model #2 Demonstration – Spokane area

As seen in the predictive model output demonstration, both of our models tend to predict home regardless of time. The reason behind this is due to the clients chosen for each of the models. For Model #1 for those living in the Portland/Vancouver area, all clients work from home. For Model #2 for those living in the Spokane area, all but one work from home. We tried to counteract this issue by forcing another cluster (second place in terms of time spent) to be the work cluster, but the model predictions remained the same.

Various evaluation metrics were used to evaluate the two predictive models. For overall accuracy, it was found that Model #1 had an overall accuracy of 78.15% and Model #2 had an overall accuracy of 80.83%. While the accuracies are not ideal for a predictive model, looking at the client-wise accuracy gives us an insight as to why this occurs.

Table 4. Client-wise Accuracy for Model #1,

Client ID	Accuracy
64	90.02%
182	88.18%
272	97.33%
273	40.92%

Table 5. Client-wise Accuracy for Model #2

Client ID	Accuracy
67	34.24%
68	82.22%
70	80.96%
328	91.84%
338	59.74%

For Model #1, the model is more accurate with predictions for Clients 64, 182 and 272. The low accuracy of Client 273 is due their frequent travel habits, within Oregon as well as regular trips to Seattle. With this in mind, we can train a refined model for individuals in Portland/Vancouver by excluding Client 273. The overall accuracy is now improved to 92.65%. For Model #2, the model is more accurate with clients 68, 70, 328. Client 67 travels frequently across the USA, leading to the lower accuracy. As for Client 338, while they do not travel out of Spokane often, they work at an office location away from home, while all the other clients work from home. We can now train a refined model for individuals in Spokane by excluding Client 67 and Client 338, which increase the overall accuracy to 82.62%. While these models are not extremely accurate, they are sufficient for the objective of this project which is to create a generalized human mobility prediction.

For further evaluation, we can study the confusion matrix of the models. Based on the matrix, we see both models always predict home, leading to high number of false positives for "home", due cases where the actual cluster is "work" or "other", but the model incorrectly predicts home [7].

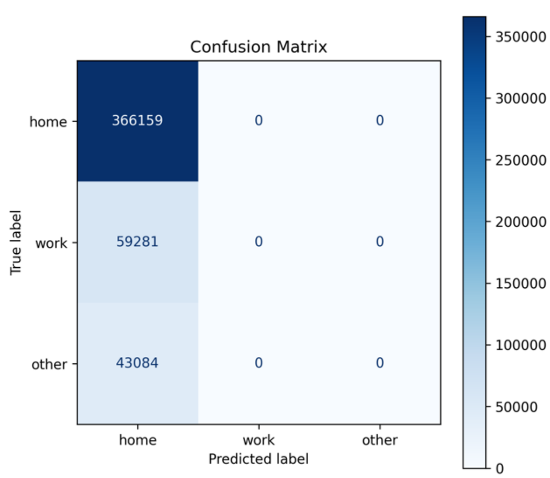

Figure 12. Model #1 Confusion Matrix

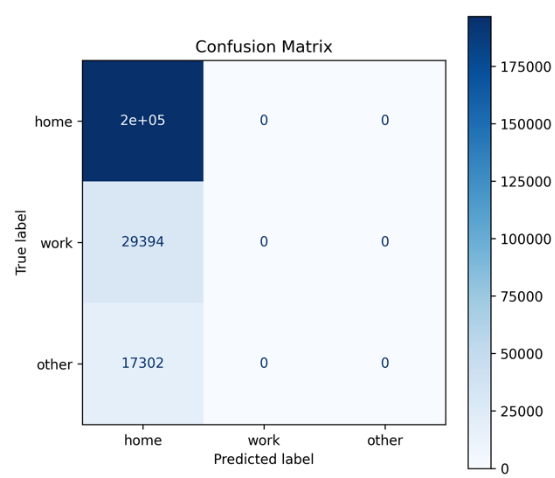

Figure 13. Model #1 Confusion Matrix

This is also mirrored by the classification report for each model. For Model #1, the precision for home is 0.78 while it is 0 for the other state labels, showing that the model is right 78% of time when it predicts home, and 22% of the time the actual state label would be work or other. For Model #2, the precision for home is 0.81 while it is 0 for the other state labels, showing that the model is right 81% of time when it predicts home, and 19% of the time the actual state label would be work or other. In both models, the recall for home is 1.00, meaning that if the actual state label is home, the model predicts home 100% of the time [8].

Model 1 Classification Report:

	precision	recall	f1-score	support
home	0.78	1.00	0.88	366159
work	0.00	0.00	0.00	59281
other	0.00	0.00	0.00	43084
accuracy			0.78	468524
macro avg	0.26	0.33	0.29	468524
weighted avg	*0.61*	*0.78*	*0.69*	*468524*

Model 2 Classification Report:

	precision	recall	f1-score	support
home	0.81	1.00	0.89	196832
work	0.00	0.00	0.00	166
other	0.00	0.00	0.00	46530
accuracy			0.81	243528
macro avg	0.27	0.33	0.30	243528
weighted avg	*0.65*	*0.81*	*0.72*	*243528*

VI. Related Work

Geospatial analysis is an extensive field with many interdisciplinary applications, one of them being human location data analysis. 'Discovering locations and habits from human mobility data' [9] uses a similar methodology to this project, using DBSCAN for clustering, along with Gaussian Mixture Model, which is also a probabilistic model like Naïve Bayes, for classifying mobility habits. The conclusions of this paper also align with our findings, by identifying a person's schedule and mobility patterns. Our project elaborates on this by determining frequent locations and estimating home and work locations. 'Simulating human mobility patterns in urban areas' [3] has a similar intuition to this project, using human movement patterns to determine how often people visit certain locations of interest, which can be used to optimize an urban space for quick access to essential resources and services. While the methodology and conclusions differ greatly from this project, this paper provides insight on future applications of this project in urban planning and resource management. 'Analysis of human mobility patterns from GPS trajectories and contextual information' [2] studies human patterns to identify travel methods as well as significant places visited. This study also looks into "third spaces" or places people tend to spend time apart from home or work. While similar research has been done, this project is distinct in that it is able to analyze each person's behavior individually by comparing heatmaps, due to the smaller size of the dataset. Additionally, the Naïve Bayes model used predicts the cluster category of where the person is likely to be at, which is unlike other research conducted in this discipline so far.

VII. Conclusion

This project analyzes the Locations dataset to observe movement patterns and identify frequent locations. It was found that people tend to spend most of their time at home, and when they venture out of their house, they visit locations within their home state. Variations in travel habits were also seen, with some individuals having rare, one-off trips across the years, while others travel regularly to various locations. The analysis is taken further by proposing a probabilistic model which can estimate a person's location using the time and day of the weekend. This model tends to veer towards predicting home for all time of the day.In the future, the dataset could be extended to improve model accuracy, by adding more attributes such as client information (such as their occupation and work hours) would allow for improved frequent location analysis to accurately estimate work locations. Additionally, with a larger, more diverse dataset, the methods presented in this paper can be used to contribute to large scale operations such as traffic monitoring predictions as well as urban planning to design city infrastructure.

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Appendix

GitHub Repository: https://github.com/trisha-tt/geospatial-analysis-project

AI Use Disclosure: GenAI was used for generating code outline and snippets.